

1 Linear projections and where they break

When is a projection just a shadow?

- PCA via SVD with `prcomp`; why not `princomp` or an eigendecomposition of the covariance.
- At $\theta = 0$ two components are exact. Variance explained collapses as θ rises.
- One curve — and it motivates the whole book.
- Kernel PCA as the bridge into Part II.